Unit 2 – Selection and Iteration Name: _____
Circuit Activity – Tracing while Loops

Beginning in cell #1, trace through the code to answer the question presented. Once you get your answer, hunt for it in the worksheet, find it, and make that problem #2. Continue in this manner until you complete the circuit.

Answer: 60

#1

What is the last number printed in the code segment below?

```
int x = 10;
while(x > 0)
{
    System.out.print(x + " ");
    x -= 2;
}
```

Answer: –1

_____

What is the first number printed in the code segment below?

```
int x = 1;
while(x < 20) {
    if (x % 5 == 0) {
        System.out.print(x + " ");
        x += 2;
    }
    else {
        x += 3;
    }

}
```

Answer: 4

_____

What is printed when the following code segment is executed?

```
int x = 20;
boolean on = true;
while(x > 0)
{
    if(on) {
        x -= 7;
    } else {
        x -= 5;
    }
    on = !on;
}
System.out.println(x);
```

Answer: 3

_____

What is printed when the code segment is run?

```
int x = 2;
while(x < 50)
{
    x *= 3;
    if(x > 20)
    {
        x /= 2;
    }
}
System.out.print(x);
```

Answer: –4

_____

What is the value of count after the code segment is done executing?

```
int x = 0;
int count = 0;
while(x < 10) {
    if(x % 2 == 0 || x % 3 == 0) {
        x += 1;
    }
    else {
        x += 2;
    }
    count++;
}
```

Answer: 10

_____

What is the last number printed in the code segment below?

```
int x = 100;
int step = 2;
while(x > 50)
{
    System.out.print(x + " ");
    x -= step;
    step += 2;
}
```

Answer: 11

_____

What is printed when the following code segment is executed?

```
int x = 1;
while(x < 20 && x > -30)
{
    x = -x * 2;
}
System.out.print(x);
```

Answer: 2

_____

What is the last number printed in the code segment below?

```
int x = 1;
while(x <= 10 && x % 7 != 0) {
    if(x % 4 == 0) {
        x += 2;
    }
    System.out.print(x + " ");
    x += 1;
}
```

Answer: 1

_____

What is the value of count after the code segment is done executing?

```
int x = 2;
int count = 0;
while(x < 50) {
    x = x * 2 + 3;
    count++;
}
```

Answer: 50

_____

What is printed when the following code segment is executed?

```
int x = 10;
while(x >= 0) {
    x -= (x / 10) + 2;
}
System.out.println(x);
```

Answer: 9

_____

What is printed when the following code segment is executed?

```
int num = 255;
int x = 1;
while(num > 0)
{
    int y = num % 10;
    x *= y;
    num /= 10;
}
System.out.println(x);
```

Answer: –32

_____

How many times does "*" get printed in the code segment below?

```
int x = 2;
while(x < 50) {
    if(x % 2 == 0) {
        x *= 3;
    }
    else {
        x += 5;
    }
    System.out.print("*");
}
```

Answer: 5

_____

What is the last number printed in the code segment below?

```
int x = 0;
while(x <= 10)
{
    x++;
    System.out.print(x + " ");
    x++;
}
```

Answer: 58

_____

What is printed when the following code segment is executed?

```
int a = 1;
int b = 4;
while(b != -1 && a % 5 > 0) {
   a++;
   b--;
}
System.out.println(a + b);
```

Answer: 6

_____

What is printed when the following code segment is executed?

```
int n = 20;
while(5 <= n)
{
  if(n % 2 == 1){
      n--;
  }
  n = n - 3;
}
System.out.print(n);
```

Answer: 7

_____

What is printed when the code segment is run?

```
int x = 1;
int y = 10;
while(x < y)
{
    x += 2;
    y -= 3;
}
System.out.println(x + y);
```